Theory of Data Visualization

Introduction

Data Visualization is the graphical representation of data using charts, graphs, and plots. It helps to understand patterns, trends, relationships, and insights from data in a visual format.

Data visualization makes complex data easy to understand and helps in better decision-making in business, research, and data analysis.

Data visualization is an important part of Data Science and Data Analytics.

Objectives of Data Visualization

The main objectives of data visualization are:

To represent data visually To identify trends and patterns To compare data values To analyze relationships between variables To detect outliers To support decision making Types of Data Visualization

1. Line Chart

A line chart is used to show trends over time or ordered data. It connects data points using lines.

Example: CGPA vs Salary, Sales over Years

1. Bar Chart

A bar chart is used to compare different categories of data.

Example: Number of placed vs unplaced students

1. Histogram

A histogram is used to show the frequency distribution of numerical data. It shows how data is distributed into different ranges (bins).

1. Scatter Plot

A scatter plot is used to show the relationship between two numerical variables. It helps to identify correlation between variables.

Example: CGPA vs Salary, Skills vs Salary

Related concept: Correlation

1. Box Plot

A box plot is used to show data distribution and detect outliers. It shows the five-number summary:

Minimum First Quartile (Q1) Median Third Quartile (Q3) Maximum

Related concept: Outlier

Advantages of Data Visualization Easy to understand data Saves time Helps in decision making Identifies trends and patterns Detects outliers and errors Useful in business and research Applications of Data Visualization

Data visualization is used in:

Business analytics Finance Healthcare Marketing Education Research Machine Learning Conclusion

Data visualization is an important technique used to represent data in graphical form. It helps to understand trends, patterns, and relationships between variables. Tools like Python, Matplotlib, and Seaborn are commonly used for data visualization. Data visualization helps organizations make better decisions based on data.

```
import seaborn as sns

import matplotlib.pyplot as plt

import pandas as pd

df=pd.read_csv("student_placement_synthetic.csv")

df.head(10)
```

```
     branch college_tier  cgpa  backlogs  coding_skills  dsa_score  \
0       ECE       Tier-3  6.70         0            7.6        4.4
1  Chemical       Tier-2  5.70         0            5.4        7.9
2        EE       Tier-2  7.19         0            5.6        6.8
3        CE       Tier-2  6.48         0            5.2        3.1
4       CSE       Tier-2  6.71         1            5.9        4.7
5       CSE       Tier-3  6.21         1            6.7        6.5
6       CSE       Tier-3  7.91         0            7.0        3.9
7        IT       Tier-3  6.95         0            6.1        6.0
8        CE       Tier-2  7.03         0            5.1        2.5
9        EE       Tier-2  7.34         3            4.7        6.2

   aptitude_score  communication_skills  ml_knowledge
system_design  \
0            49.5                   3.7           6.4            0.3

1            72.0                   8.3           6.3            1.9

2            79.1                   7.4           4.4            5.2

3            48.4                   5.0           1.1            6.7

4            61.2                   4.3           2.7            2.8

5            69.9                   5.0           4.0            5.2

6            61.5                   6.7           4.9            2.2
```

```
7            44.4                    5.5           4.5            6.3

8            54.7                    6.4           6.2            3.2

9            78.8                    5.9           6.7            5.2


   internships  projects_count  certifications  hackathons  \
0            1               4               4           3
1            0               4               0           0
2            1               3               2           1
3            1               4               3           0
4            1               2               0           3
5            2               3               3           2
6            2               5               2           1
7            1               2               1           1
8            1               3               0           3
9            1               2               1           1

   open_source_contributions  extracurriculars  placement_status  \
0                          2                 1                 1
1                          0                 0                 0
2                          2                 0                 1
3                          0                 0                 0
4                          0                 1                 1
5                          0                 1                 1
6                          0                 3                 1
7                          1                 0                 0
8                          0                 1                 1
9                          0                 0                 1

   salary_package_lpa
0               14.75
1                 NaN
2               19.06
3                 NaN
4               13.42
5               16.47
6               16.51
7                 NaN
8               13.49
9               17.95
```

```
df.isnull().sum()
```

```
branch                            0
college_tier                      0
cgpa                              0
backlogs                          0
coding_skills                     0
```

```
dsa_score                          0
aptitude_score                     0
communication_skills               0
ml_knowledge                       0
system_design                      0
internships                        0
projects_count                     0
certifications                     0
hackathons                         0
open_source_contributions          0
extracurriculars                   0
placement_status                   0
salary_package_lpa             31525
dtype: int64
```

```
df["salary_package_lpa"].value_counts()
```

```
salary_package_lpa
17.57    128
17.78    125
16.47    124
16.80    122
15.87    122
        ...
9.11       1
24.56      1
10.71      1
8.65       1
26.11      1
Name: count, Length: 1654, dtype: int64
```

```
df.isnull().sum()
```

```
branch                             0
college_tier                       0
cgpa                               0
backlogs                           0
coding_skills                      0
dsa_score                          0
aptitude_score                     0
communication_skills               0
ml_knowledge                       0
system_design                      0
internships                        0
projects_count                     0
certifications                     0
hackathons                         0
open_source_contributions          0
extracurriculars                   0
placement_status                   0
```

```
salary_package_lpa            31525
dtype: int64
```

```
df["salary_package_lpa"].median()
```

```
np.float64(17.25)
```

```
df.isnull().sum()
```

```
branch                            0
college_tier                      0
cgpa                              0
backlogs                          0
coding_skills                     0
dsa_score                         0
aptitude_score                    0
communication_skills              0
ml_knowledge                      0
system_design                     0
internships                       0
projects_count                    0
certifications                    0
hackathons                        0
open_source_contributions         0
extracurriculars                  0
placement_status                  0
salary_package_lpa            31525
new_Package                       0
dtype: int64
```

```
df["salary_package_lpa"] = df["salary_package_lpa"].astype(float)
```

```
df["salary_package_lpa"] = df["salary_package_lpa"].fillna(17.02)
```

```
df.isnull().sum()
```

```
branch                        0
college_tier                  0
cgpa                          0
backlogs                      0
coding_skills                 0
dsa_score                     0
aptitude_score                0
communication_skills          0
ml_knowledge                  0
system_design                 0
internships                   0
projects_count                0
certifications                0
hackathons                    0
open_source_contributions     0
```

```
extracurriculars              0
placement_status              0
salary_package_lpa            0
new_Package                   0
dtype: int64
```

```
df.head(10)
```

```
     branch college_tier  cgpa  backlogs  coding_skills  dsa_score  \
0       ECE       Tier-3  6.70         0            7.6        4.4
1  Chemical       Tier-2  5.70         0            5.4        7.9
2        EE       Tier-2  7.19         0            5.6        6.8
3        CE       Tier-2  6.48         0            5.2        3.1
4       CSE       Tier-2  6.71         1            5.9        4.7
5       CSE       Tier-3  6.21         1            6.7        6.5
6       CSE       Tier-3  7.91         0            7.0        3.9
7        IT       Tier-3  6.95         0            6.1        6.0
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   aptitude_score  communication_skills  ml_knowledge
system_design  \
0            49.5                   3.7           6.4            0.3

1            72.0                   8.3           6.3            1.9

2            79.1                   7.4           4.4            5.2

3            48.4                   5.0           1.1            6.7

4            61.2                   4.3           2.7            2.8

5            69.9                   5.0           4.0            5.2

6            61.5                   6.7           4.9            2.2

7            44.4                   5.5           4.5            6.3

8            54.7                   6.4           6.2            3.2

9            78.8                   5.9           6.7            5.2


   internships  projects_count  certifications  hackathons  \
0            1               4               4           3
1            0               4               0           0
2            1               3               2           1
3            1               4               3           0
4            1               2               0           3
5            2               3               3           2
6            2               5               2           1
```
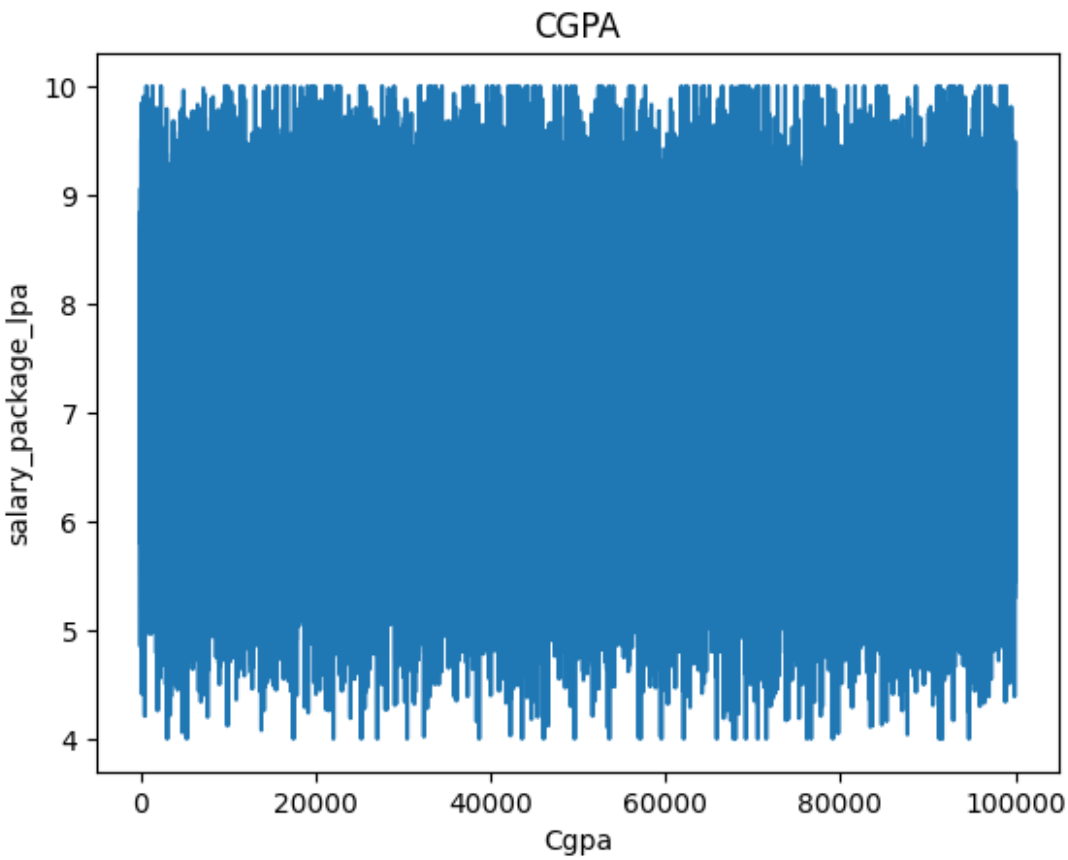

```
7            1                2                1            1
8            1                3                0            3
9            1                2                1            1

   open_source_contributions  extracurriculars  placement_status  \
0                          2                 1                 1
1                          0                 0                 0
2                          2                 0                 1
3                          0                 0                 0
4                          0                 1                 1
5                          0                 1                 1
6                          0                 3                 1
7                          1                 0                 0
8                          0                 1                 1
9                          0                 0                 1

   salary_package_lpa  new_Package
0               14.75        14.75
1                0.00        17.25
2               19.06        19.06
3                0.00        17.25
4               13.42        13.42
5               16.47        16.47
6               16.51        16.51
7                0.00        17.25
8               13.49        13.49
9               17.95        17.95
```

```
df.columns
```

```
Index(['branch', 'college_tier', 'cgpa', 'backlogs', 'coding_skills',
       'dsa_score', 'aptitude_score', 'communication_skills',
'ml_knowledge',
       'system_design', 'internships', 'projects_count',
'certifications',
       'hackathons', 'open_source_contributions', 'extracurriculars',
       'placement_status', 'salary_package_lpa', 'new_Package'],
      dtype='str')
```

Line plot

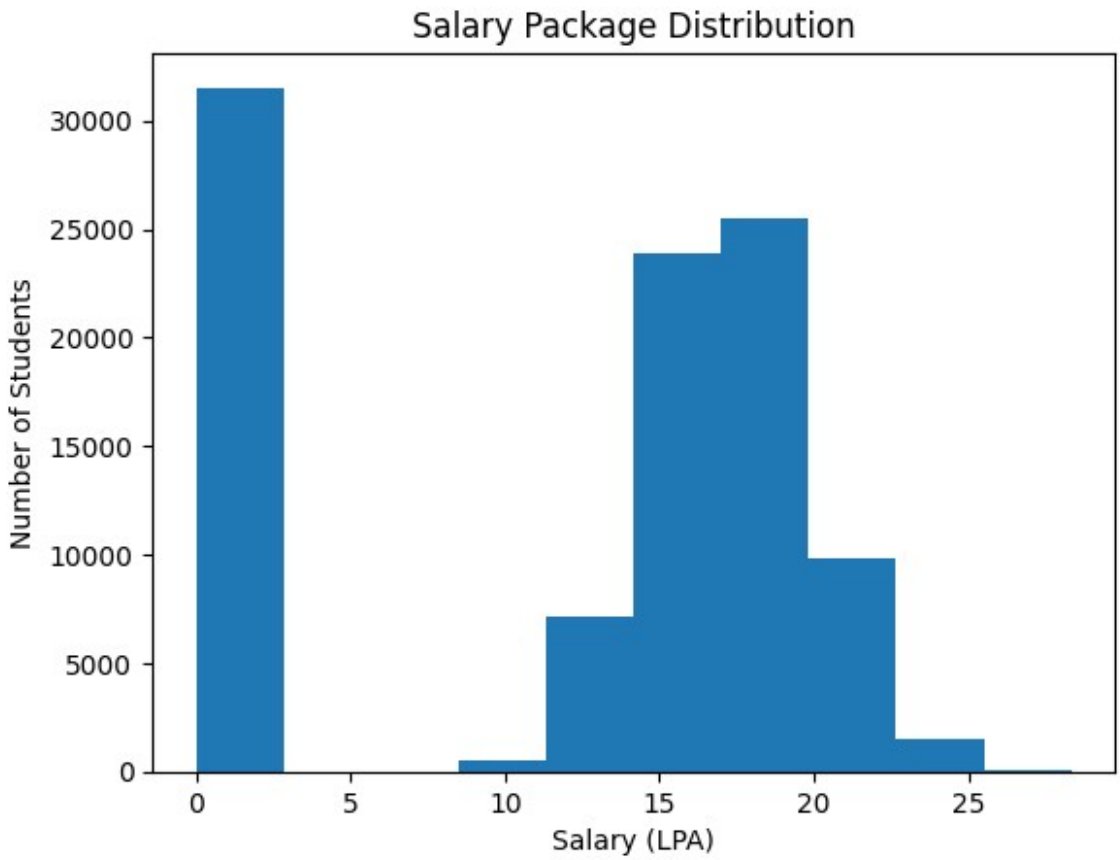
```
plt.plot(df["cgpa"])
plt.xlabel("Cgpa")
plt.ylabel("salary_package_lpa")
plt.title("CGPA")
plt.show()
```



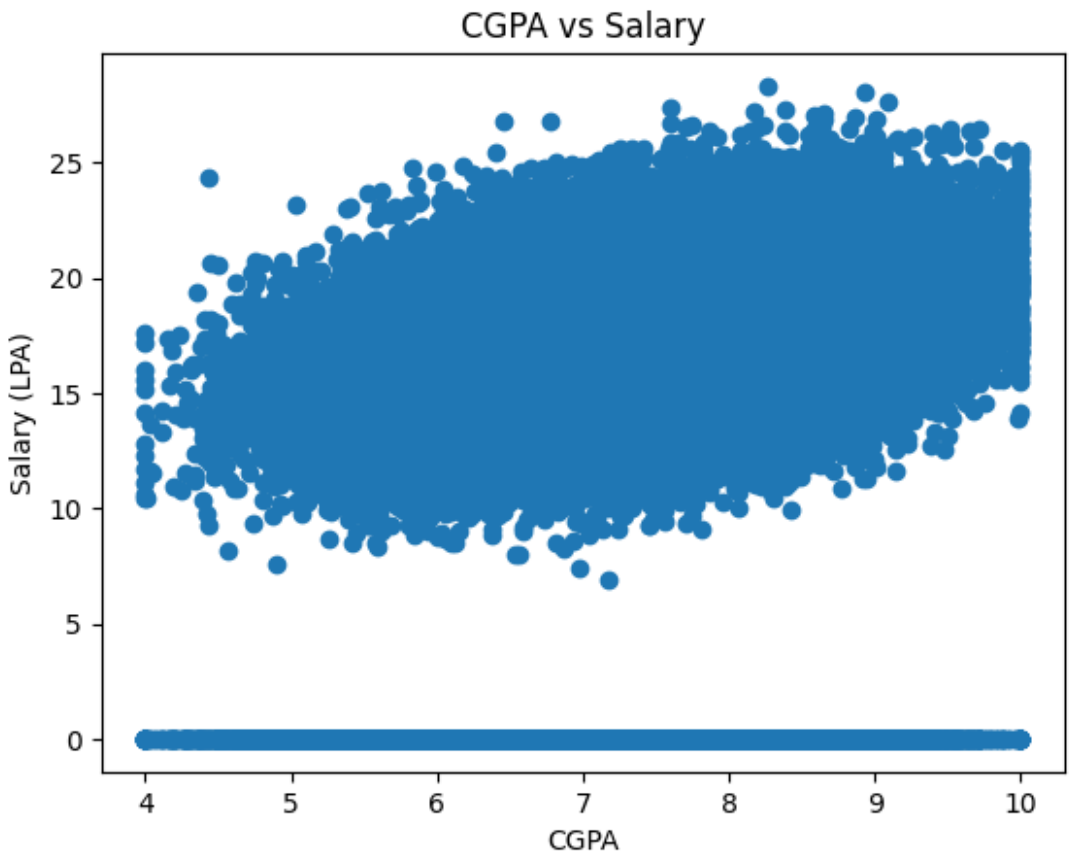

```
plt.plot(df["projects_count"], df["salary_package_lpa"])
plt.xlabel("Projects Count")
plt.ylabel("Salary")
plt.title("Projects vs Salary")
plt.show()
```

Histogram

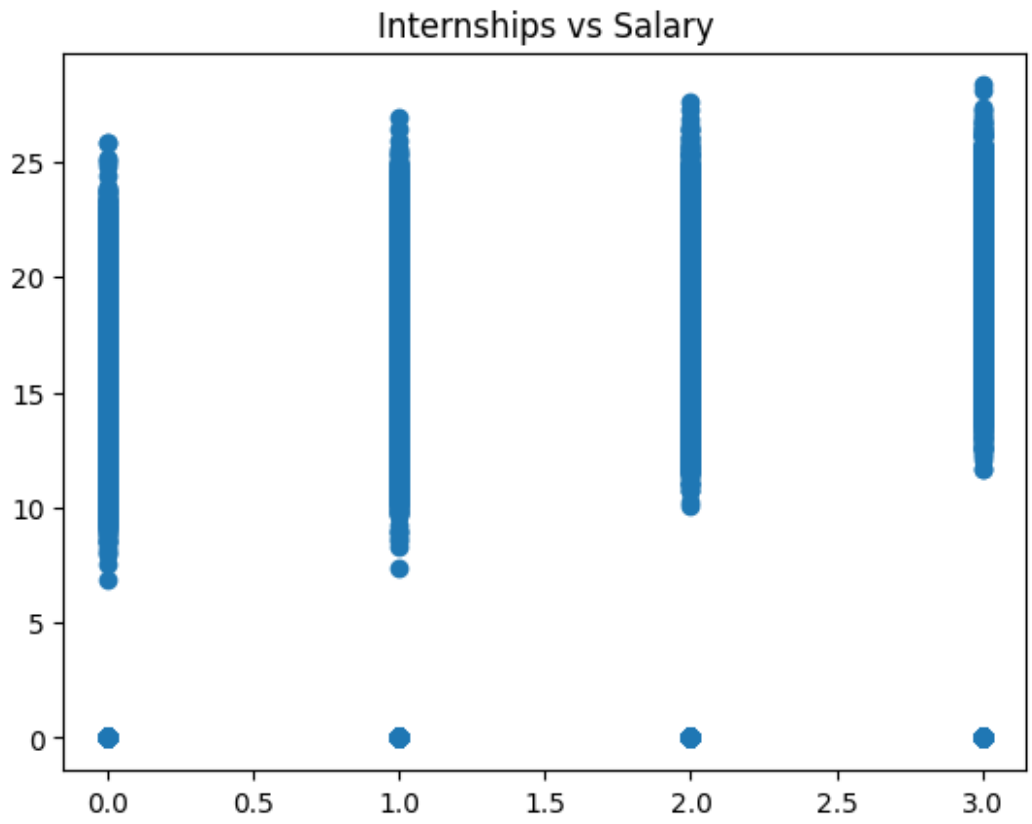
```
plt.hist(df["salary_package_lpa"])
plt.title("Salary Package Distribution")
plt.xlabel("Salary (LPA)")
plt.ylabel("Number of Students")
plt.show()
```



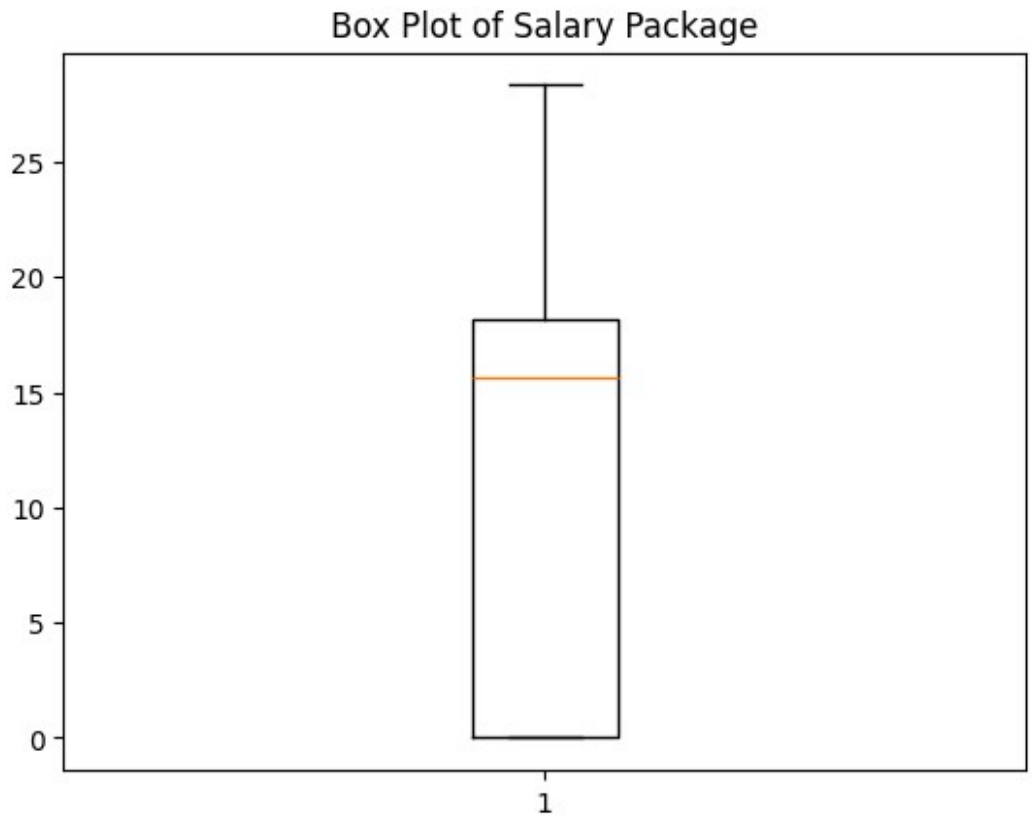

```
plt.scatter(df["cgpa"], df["salary_package_lpa"])
plt.title("CGPA vs Salary")
plt.xlabel("CGPA")
plt.ylabel("Salary (LPA)")
plt.show()
```




```
plt.scatter(df["internships"], df["salary_package_lpa"])
plt.title("Internships vs Salary")
plt.show()
```




```
plt.boxplot(df["salary_package_lpa"])
plt.title("Box Plot of Salary Package")
plt.show()
```


Box Plot of Salary Package
25
20
15
10
5
0
1